

MATE 481/681 FUNDAMENTALS OF WELDING

Syllabus and Calendar
Fall 2021



MatE 481: Fundamentals of Welding Engineering

MatE 681: Advanced Welding Engineering

Welcome to MATE 481/681. This is an engineering course in which we aim at introducing students to the fundamental aspects of welding engineering in the best possible way: in front of the actual equipment. The course has a significant lab demonstration component.

After completing this course students should master the concepts related to a vast variety of welding processes, the laws of nature on which they operate, related materials issues, and the practical implications of the new concepts learned, including the writing of a welding procedure.

The TAs and I have prepared this course with much love and effort. This course has been much appreciated by students over many years now. We hope that in addition to learning much, you'll also have an enjoyable experience.

Thank you for choosing to take this course!

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Schedule

Lectures: Tuesdays, Thursdays 2:00 – 3:20 NRE 2-090
Seminar Tuesdays 3:30 – 4:20 NRE 2-090
Office hours: By appointment

The lectures will consist of a combination of theory and worked examples. Lectures will involve significant use of the whiteboard; please make sure you bring writing materials to take notes. On occasion the lecture time will be used for demos or hands-on experience at the lab.

The seminars will include practice time at the lab, lectures by guests from industry, and practical problem solving.

Teaching Assistants

Eduardo Alvarez: jealvare@ualberta.ca
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Class Material

The eClass site contains:

- Class announcements
- Syllabus and schedule
- Class Topics
 - Slides
 - Supplemental material
 - Videos of 2019 class (for most, but not all topics)
 - Videos of 2020 class (all topics)
- Most previous exams with answer keys (except 2020)
 - **Important note:** This class relies much on current research, and has evolved in time as newer material is included. The answer key to old exams might need to be reinterpreted using the updated values, concepts, and notation. In case of inconsistency, the concepts taught this year prevail over those of previous years.
- Instructions for the Assignment and Project.

Textbooks

Mandatory: The Procedure Handbook of Arc Welding (US \$30)

<http://www.jflf.org/ProductDetails.asp?ProductCode=PH>

You can purchase the book online individually, but the cost of shipping is very high. TAs have arranged to have some books already available to offset the shipping costs.

Recommended: AWS Welding Handbook, Volume 1: Welding Science and Technology

Available in Knovel for free for UofA students

Evaluation

Quiz 1:	Tuesday, Sep. 14, 3:30-4:20, in class
Quiz 2:	Tuesday, Oct. 12, 3:30-4:20, in class
Quiz 3:	Tuesday, Nov. 2, 3:30-4:20, in class
Quiz 4:	Tuesday, Nov. 23, 3:30-4:20, in class
Assignment:	Monday Dec. 13, 5:00 submitted via eClass
Project (only 681):	Monday Dec. 13, 5:00 submitted via eClass

MatE 481:

Quiz 1:	10%
Quiz 2:	15%
Quiz 3:	25%
Quiz 4:	25%
Assignment:	25%

MatE 681:

Quiz 1:	8%
Quiz 2:	12%
Quiz 3:	20%
Quiz 4:	20%
Assignment:	20%
Project:	20%

Topic of projects assigned: by Tuesday, Sep. 28

All exams, assignment, and projects are individual, not group efforts. Discussion is encouraged for the assignment and projects, but it is not permitted during the exams.

COVID Considerations

In-person attendance:

Lectures and seminars

We will abide to UofA protocols. I expect to deliver lectures without a mask 2m away from the front row. Attendants must wear a mask during class, except UofA approved exceptions.

Lab demos

On selected dates I intend to host lab demos where you will see in person the welding machines in action. If possible within the safety requirements, students will also have hands-on welding experience.

Quizzes

Quizzes will be in NRE 2-090 and will follow standard written exam practice. Topics for each quiz, as well as details such as open or closed book modality will be detailed before each exam in eClass.

Assignment and Report

Submitted online through eClass

Remote attendance:

Lectures and seminars

Delivered asynchronously (i.e. pre-recorded). Pre-recorded videos feature the same content than the in-person class. Location of videos is in the corresponding topics of eClass.

Lab demos

Lab demos have been pre-recorded and videos are in eClass, together with their corresponding lectures.

Quizzes

Quizzes will be taken remotely. The exact mode and time of the quiz, as well as topic and details such as open or closed book modality will be detailed before the exam in eClass.

Assignment and Report

Submitted online through eClass

MATE 481/681 Tentative Calendar

Date	Topic	Seminar
2021-09-02 Thu	Intro to welding	
2021-09-07 Tue	Concept of Joining	Welding Safety
2021-09-09 Thu	Terminology, Welding symbols	
2021-09-14 Tue	Welding processes	Quiz 1
2021-09-16 Thu	Welding processes	
2021-09-21 Tue	Welding processes	Demo GTAW, PTAW
2021-09-23 Thu	Welding processes	
2021-09-28 Tue	Welding processes	Demo GMAW /FCAW/SAW
2021-09-30 Thu	Holiday	
2021-10-05 Tue	Consumable naming, Fluxes	Demo SMAW
2021-10-07 Thu	Determining procedure from book	
2021-10-12 Tue	Welding Metallurgy, C steel and CCTs, Ceq	Quiz 2
2021-10-14 Thu	Welding Metallurgy, C steel and CCTs, Ceq	
2021-10-19 Tue	Welding Metallurgy, C steel and CCTs, Ceq	Prediction of hardness
2021-10-21 Thu	The welding arc	
2021-10-26 Tue	The welding arc, Voltage in arc welding	Voltage, arc
2021-10-28 Thu	Energy balance in arc welding	
2021-11-02 Tue	Deposition in arc welding	Quiz 3
2021-11-04 Thu	Mass balance	
2021-11-09 Tue	Reading week	
2021-11-11 Thu	Reading week	
2021-11-16 Tue	Moving heat sources	Deposition, temperature
2021-11-18 Thu	Moving heat sources, Weld Pool	
2021-11-23 Tue	Weld Design	Quiz 4
2021-11-25 Thu	Weld Design	
2021-11-30 Tue	Design rules for welds (by Prof. Duane Miller)	Codes and Standards (Dr. N. Barnes)
2021-12-02 Thu	Codes and Standards (Kurtis Bell)	
2021-12-07 Tue	NDT techniques (Dr. Eric Sjerpe)	NDT demo (IrisNDT @ CCWJ lab)
2021-12-13 Mon	Assignment, Project due	